IWEE: Inverse probability Weighted Estimating Equations for microbiome analyses

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Abstract

Zero-inflation is a common problem in microbiome studies where typically 50% or more of the microbiome sequencing reads could be zero. Many existing methods require imputing zeros with a positive number, such as a Pseudocount, to make inference. However, imputation approaches are problematic because there is no consensus on which values to use to replace zeros, which can distort the true underlying associations. In this paper, we propose a novel approach to address this issue by employing inverse probability weighting so that data points with a higher probability of being observed as zeros receive more weight to reduce bias in the analysis. This new approach can also handle the compositional structure of microbiome data and make inference on the absolute abundance of an ecosystem. We conduct an extensive simulation study to demonstrate the proposed method's performance compared with existing approaches. Two real-world applications are presented to compare the method with existing approaches.

1 Introduction

The human microbiota consists of 10-100 trillion symbiotic cells residing in each person. It has been widely reported that the human microbiome plays an important role in human health (HMPConsortium (2012), Kau et al. (2011), Lloyd-Price et al. (2017), Turnbaugh et al. (2007)) and is associated with a variety of diseases including diabetes (Komaroff (2017)), Alzheimer (Wang et al. (2019)), and cancers (Castellarin et al. (2012), Reikvam et al. (2011)). 16S RNA sequencing (Cole et al. (2009)) and metagenomic shotgun (Tringe and Rubin (2005)) are commonly used to produce sequencing reads for microbiome data. In human studies, the microbiome data are

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typically acquired from microbiota in samples (e.g., stool and saliva) to infer the associations of interest within the ecosystem (e.g., gut, oral cavity). Collected samples represent only a small fraction of the ecosystem, so their raw sequencing counts (i.e., absolute abundance (AA)) do not directly estimate the AA in the ecosystem. In particular, the proportion of the ecosystem represented by the collected sample (i.e., sequencing depth) is usually unknown. Researchers therefore prefer to analyze relative abundance (RA), which represents the fraction of taxa within the ecosystem, and can be approximated using the RA observed from the collected samples (Chen and Li (2013), La Rosa et al. (2012), Lozupone and Knight (2005), Tang and Chen (2018)).

A well-known issue in microbiome data is the excessive number of zeros in sequencing reads. It is common to see 50% or more of the sequencing reads being zero. These zeros may arise for different reasons. For example, an issue in the data-processing pipeline may cause reads to be missed; we refer to these as mechanical zeros. Alternatively, a taxon's abundance may be too low for detection; we refer to these as floor zeros. Current technologies generally cannot distinguish between these two types of zeros. The zero-inflation feature makes it challenging to analyze microbiome data as a log-transformation is usually needed for microbiome data analysis due to its skewness, but the log-transformation cannot be applied to a zero. Many existing methods such as ANCOM-BC2 (Lin and Peddada (2020)), metagenomeSeq (Paulson et al. (2013)), CLR transformation (Zhou et al. (2022)), address this issue by imputing zeros with positive numbers before the analysis. However, there is no consensus on an appropriate replacement value. Pseudocount has been suggested for AA data, whereas different imputation values have been used for RA data, yielding results that conflict with the pseudocount approach (Zhou et al. (2022)). Moreover, as pointed out in Li et al. (2021), imputed values could distort the associations of interest because imputation can drastically change the true values on a log-scale, and lead to artificial bias.

To address zero inflation, we propose a novel inverse-probability-weighted estimating equation (IWEE) method. It models the probability that an observation is a mechanical zero and accommodates data containing both mechanical and floor zeros. Each observation is weighted by the inverse of its estimated probability of being a mechanical zero, so observations more likely to be mechanical zeros receive greater weight. This approach follows the principle of inverse probability weighting for missing data, in which observations are weighted by the inverse of their probability of being observed. IWEE accounts for inter-taxon correlations and, through the use of robust estimating equations, remains robust to misspecification of the correlation structure. It also accommodates the compositional nature of microbiome data and enables inference on the AA of an ecosystem. To support valid inference, it incorporates the IFAA approach Li et al. (2021), which identifies a suitable reference taxon independent of the covariates of primary interest.

We further evaluated the proposed method using data from the Marijuana Associated Planning and Long-term Effects (MAPLE) study, a longitudinal prospective cohort designed to investigate the health effects of cannabis use among people living with HIV in Florida (Porchia et al. (2025)). The analysis used data from the MAPLE microbiome and Alzheimer's supplemental substudy, which included older adults with HIV and their microbiome measurements. Although the par-

ent MAPLE study is longitudinal, this microbiome analysis was cross-sectional and focused on baseline cannabis exposure and gut microbial composition. The original study examined whether cannabis use was associated with gut microbial diversity and genus-level abundance, motivated by the potential effects of cannabinoids on intestinal inflammation, mucosal barrier integrity, and microbial composition. In our analysis, we used genus-level absolute abundance as the outcome, baseline marijuana consumption in grams per day as the primary exposure, and participant age and gender as potential confounders. After quality control and filtering, 248 genera were retained. This application illustrates the use of IWEE in a regression setting with a continuous exposure and highly sparse microbiome data.

To demonstrate the applicability of IWEE in a randomized clinical trial setting, we also analyzed data from the Liberalized Diet and Neutropenic Diet Study (Wingard et al. (2026)), a phase III randomized noninferiority trial involving adults undergoing hematopoietic stem-cell transplantation or intensive induction therapy for acute leukemia or myelodysplastic syndrome. The trial compared a liberalized diet, which permitted the consumption of fresh fruits and vegetables, with a traditional neutropenic diet during periods of neutropenia. Its primary objective was to determine whether the liberalized diet was noninferior to the neutropenic diet with respect to major infections; secondary objectives included evaluating caloric intake, symptoms, and quality of life. The original trial found that the liberalized-diet group experienced more major infections and did not demonstrate improvements in caloric intake, symptoms, or quality of life. In our analysis, we focused on microbiome data collected at the end of the intervention to identify how different dietary habits affect the composition of the gut microbiota. After quality control and data preprocessing, the dataset included 88 participants and 304 bacterial genera. We treated randomized diet assignment as the primary exposure and applied IWEE to evaluate its association with genus-level microbiome composition. This application demonstrates IWEE in a differential-abundance setting with a binary randomized exposure and no additional adjustment covariates.

The article is organized as follows. The model and notations are introduced in section 2, followed by a description of zero mechanisms in section 3. Parameter estimation is illustrated in section 4. We then showcase the performance of our method in comparison with a commonly used existing method by extensive simulations in section 5, as well as two real data applications in section 6, followed by discussion in section 7.

2 Model and Notation

Suppose there are N subjects and $K+1$ taxa of interest. Let $\mathcal{Y}_i = (\mathcal{Y}_i^1, \mathcal{Y}_i^2, ..., \mathcal{Y}_i^K, \mathcal{Y}_i^{K+1})^T$ denote the true microbial taxa absolute abundance (i.e., counts, AA) in the ecosystem (e.g., gut) of the ith subject, $i = 1, ..., N$. Oftentimes, the study goal is to investigate the associations of microbiome abundance with covariates that could change microbiome composition. Suppose there are Q covariates of interest, denoted by the Q-dimensional vector X_i, for the association test. X could be the treatment variable in a controlled trial setting or an exposure variable in an

observational study setting. Let's say there are other S covariates as confounders in the model. These potential confounders, denoted by a S-dimensional vector W_i, are controlled in the model, but their associations with the microbiome will not be examined. In this paper, we focus on the association between X_i and $\mathcal{Y}_i$ adjusting for potential confounders W_i. We use the following equations to model the marginal associations:

$$E\big(\log(\mathcal{Y}_i^k)\big) = \beta^{0k} + X_i^T\beta^k + W_i^T\gamma^k, \;\; i = 1,\ldots,N; k = 1,\ldots,K+1, \tag{1}$$

The parameter vector β^k quantifies average change in the abundance of taxon k on log scale given one unit change in covariates contained in X_i.

3 Zeros in the data

In real studies, the true taxa abundances, $\mathcal{Y}_i$, in an ecosystem usually cannot be observed because only a small portion of the ecosystem (e.g., stool sample) is used to produce sequencing data. What can be observed for the ith subject and kth taxon is $Y_i^k = C_i\mathcal{Y}_i^k$, where C_i is the unknown proportion and takes a value between 0 and 1. It is straightforward to see that C_i is directly related to sequencing depth such as library size. Let $R_i = (R_i^1, ..., R_i^{K+1})^T$ denote the vector of binary variables indicating whether Y_i^k are observed or set to be zero. When $R_i^k = 1$, we observe a positive value for Y_i^k. When $R_i^k = 0$, we observe 0 for Y_i^k. We consider a mixture of two types of zeros with the following probabilities:

$$\begin{cases} \text{Mechanical zero:} \quad P(R_i^k = 0 \mid Y_i^k) = \exp(-\eta Y_i^k), & \text{if } Y_i^k \geq 1, \quad (2a) \\ \text{Floor zero:} \quad P(R_i^k = 0 \mid Y_i^k) = 1, & \text{if } Y_i^k < 1. \quad (2b) \end{cases}$$

where η is a positive parameter that needs to be determined.

Under these two mechanisms, smaller values of Y_i^k are more likely to be observed as zero since $exp(-\eta Y_i^k)$ is a decreasing function of Y_i^k. We refer to these zeros as "mechanical zero". And, when Y_i^k is less than 1, it is always observed as zero, which is referred to as "floor zero" in this paper.

4 Parameter estimation

We are interested in estimating the parameters $\beta^k, k = 1,\ldots,K+1$, that quantify the associations of the exposure variables and microbiome in equation (1). Notice that the latent variable C_i masks the true value of $\mathcal{Y}_i^k$. Without appropriately accounting for the impact of C_i, the estimate of β^k could be distorted toward overestimation, leading a high false positive rate, or underestimation, which leads to a high false negative rate. Our method can efficiently address this issue.

4.1 Known reference taxon

Our approach involves identifying an optimal reference taxon independent of the covariates of interest. For illustration purposes, let's first assume that we know there is a taxon independent of X_i and it is set to be the reference taxon. Without loss of generality, we label this reference taxon as $K+1$ for which $\beta^{K+1}=0$. We will explain the case with an unknown reference taxon in a later section. By taking the log-ratio of RA of a taxon, say taxon k, over the reference taxon, we have:

$$\begin{aligned}
&E\big(\log(Y_i^k/Y_i^{K+1})\big)\\
&=E\big(\log(Y_i^k)-\log(Y_i^{K+1})\big)\\
&=E\Big(\log(C_i)+\log(\mathcal{Y}_i^k)-\log(C_i)-\log(\mathcal{Y}_i^{K+1})\Big)\\
&=E\big(\log(\mathcal{Y}_i^k)\big)-E\big(\log(\mathcal{Y}_i^{K+1})\big)\\
&=\beta^{0k}-\beta^{0,K+1}+X_i^T(\beta^k-\beta^{K+1})+W_i^T(\gamma^k-\gamma^{K+1})\\
&=\beta^{0k}-\beta^{0,K+1}+X_i^T\beta^k+W_i^T(\gamma^k-\gamma^{K+1})\\
&=Z_i^T\alpha^{k,K+1},
\end{aligned}\tag{3}$$

where $Z_i=(1,X_i^T,W_i^T)^T$ and $\alpha^{k,K+1}=\Big(\beta^{0k}-\beta^{0,K+1},\beta^{k^T},(\gamma^k-\gamma^{K+1})^T\Big)^T$ which can be considered as a new parameter vector by re-parameterization. Notice the term $\log(C_i)$ is canceled out in equation (3), and thus its impact is limited. Following (Preisser et al. (2002)), we can construct a weighted estimating equation to estimate the vectors $\alpha^{k,K+1}$, from which the corresponding β^k can be obtained. Given the conditional expectation of $\log(Y_i^k/Y_i^{K+1})$ specified in equation (3), the estimating equation takes the form:

$$\sum_{i=1}^{N}\sum_{k=1}^{K}\frac{R_i^kR_i^{K+1}\phi_{k,K+1}^{-1}}{P(R_i^k=1,R_i^{K+1}=1|Y_i^k,Y_i^{K+1})}Z_i\big(\log(Y_i^k/Y_i^{K+1})-Z_i^T\alpha^{k,K+1}\big)=0,\tag{4}$$

Under the missing-at-random assumption, solving this equation yields consistent and asymptotically normal estimators of $\alpha^{k,K+1}$, $k=1,\ldots,K$ (Robins et al. (1995)). These properties are guaranteed by the specification of equation (3) regardless of the correlation structure among samples. Although equation (4) is a valid estimating equation that can be solved to find consistent estimators of $\alpha^{k,K+1},k=1,\ldots,K$, it is inefficient because samples without sequencing data on Y_i^{K+1} do not contribute to the left side of the equation. Let $\alpha^{K+1}=((\alpha^{1,K+1})^T,\cdots,(\alpha^{K,K+1})^T)^T$. Another more general and efficient estimating equation can be constructed as follows:

$$\sum_{i=1}^{N}\sum_{1\le k_1<k_2\le K+1}\frac{R_i^{k_1}R_i^{k_2}\big(A^{k_1k_2}\mathcal{Z}_i\big)^T\phi_{k_1k_2}^{-1}\Big(U_i^{k_1k_2}-A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big)}{P(R_i^{k_1}=1,R_i^{k_2}=1|Y_i^{k_1},Y_i^{k_2})}=0.\tag{5}$$

where $U_i^{k_1,k_2}=\log(Y_i^{k_1}/Y_i^{k_2})$, $\phi_{k_1k_2}$ is the variance of $U_i^{k_1k_2}$ and $A^{k_1k_2}$ is a $1\times K$ vector with the k_1th element equal to 1, the k_2th element equal to -1, and all other elements equal to 0.

When $k_2 = K+1$, the k_2th element of $A^{k_1k_2}$ equals to 0. And $\mathcal{Z}_i = I_K \otimes (Z_i)^T$, where I_K is the $K \times K$ identity matrix and $\otimes$ denotes Kronecker product. Equation (5) involves evaluating $NK(K+1)/2$ terms, many of which might be 0 depending on the sparsity level of the data. We show in the Appendix that Equation (5) is a valid estimating equation for α^{K+1}.

4.1.1 Solution for the weighted estimating equations

According to the probability mechanism in equation (2a) for observing mechanical zero's, we have

$$\begin{aligned}
&P(R_i^{k_1} = 1, R_i^{k_2} = 1 | Y_i^{k_1}, Y_i^{k_1}) \\
&= P(R_i^{k_1} = 1 | Y_i^{k_1}) P(R_i^{k_2} = 1 | Y_i^{k_2}) \\
&= \big(1 - \exp(-\eta Y_i^{k_1})\big)\big(1 - \exp(-\eta Y_i^{k_2})\big)
\end{aligned}$$

Equation (5) becomes

$$\sum_{i=1}^{N} \sum_{1 \le k_1 < k_2 \le K+1} \frac{R_i^{k_1} R_i^{k_2} \big(A^{k_1k_2}\mathcal{Z}_i\big)^T \phi_{k_1k_2}^{-1}}{\big(1 - \exp(-\eta Y_i^{k_1})\big)\big(1 - \exp(-\eta Y_i^{k_2})\big)} \Big(U_i^{k_1k_2} - A^{k_1k_2}\mathcal{Z}_i \alpha^{K+1}\Big) = 0. \tag{6}$$

It can be solved by an iterative procedure. Let $\phi = (\phi_{12}, \phi_{13}, \cdots, \phi_{K,K+1})$ and $\phi^0 = (\phi_{12}^0, \phi_{13}^0, \cdots, \phi_{K,K+1}^0)$ denote its initial value. Let $\mathcal{S}_i = \{(k_1, k_2) : R_i^{k_1} R_i^{k_2} = 1\}$ be the set composed of all pairs of $(k_1, k_2), 1 \le k_1 < k_2 \le K+1$, which satisfies $R_i^{k_1} R_i^{k_2} = 1$ for the ith subject, and with its cardinality denoted by $|\mathcal{S}_i|$. Equation (6) can be written as

$$\begin{aligned}
&\sum_{i=1}^{N} \sum_{(k_1,k_2)\in\mathcal{S}_i} W_i^{k_1k_2} \big(A^{k_1k_2}\mathcal{Z}_i\big)^T \Big(U_i^{k_1k_2} - A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big) \\
=&\sum_{i=1}^{N} \sum_{(k_1,k_2)\in\mathcal{S}_i} \sqrt{W_i^{k_1k_2}} \big(A^{k_1k_2}\mathcal{Z}_i\big)^T \Big(\sqrt{W_i^{k_1k_2}} U_i^{k_1k_2} - \sqrt{W_i^{k_1k_2}} A^{k_1k_2}\mathcal{Z}_i\alpha^{K+1}\Big) \\
=&\sum_{i=1}^{N} \sum_{(k_1,k_2)\in\mathcal{S}_i} \tilde{\mathcal{Z}}_i^{k_1k_2} \Big(\tilde{U}_i^{k_1k_2} - \tilde{\mathcal{Z}}_i^{k_1k_2}\alpha^{K+1}\Big) \\
=&\sum_{i=1}^{N} \tilde{\mathcal{Z}}_i^T \Big(\tilde{U}_i - \tilde{\mathcal{Z}}_i\alpha^{K+1}\Big) \\
=&-\frac{1}{2}\frac{\partial l}{\partial \alpha^{K+1}} = 0,
\end{aligned}$$

where

$$\begin{aligned}
W_i^{k_1k_2} &= \frac{R_i^{k_1} R_i^{k_2} \phi_{k_1k_2}^{-1}}{\Big(1 - \exp(-\eta Y_i^{k_1})\Big)\Big(1 - \exp(-\eta Y_i^{k_2})\Big)}, \\
\tilde{\mathcal{Z}}_i^{k_1k_2} &= \sqrt{W_i^{k_1k_2}} A^{k_1k_2}\mathcal{Z}_i, \\
\tilde{U}_i^{k_1k_2} &= \sqrt{W_i^{k_1k_2}} U_i^{k_1k_2},
\end{aligned}$$

$$\tilde{U}_i = \left(\sqrt{W_i^{(1)(2)}}U_i^{(1)(2)}, \sqrt{W_i^{(1)(3)}}U_i^{(1)(3)}, \cdots, \sqrt{W_i^{(|S_i|-1)(|S_i|)}}U_i^{(|S_i|-1)(|S_i|)}\right)^T,$$

$$\tilde{\mathcal{Z}}_i = \begin{pmatrix} \sqrt{W_i^{(1)(2)}}A^{(1)(2)}\mathcal{Z}_i \\ \sqrt{W_i^{(1)(3)}}A^{(1)(3)}\mathcal{Z}_i \\ \vdots \\ \sqrt{W_i^{(|S_i|-1)(|S_i|)}}A^{(|S_i|-1)(|S_i|)}\mathcal{Z}_i \end{pmatrix},$$

$$l = \sum_{i=1}^{N} ||\tilde{U}_i - \tilde{\mathcal{Z}}_i\alpha_1^{K+1}||_2^2.$$

where $(k_1)(k_2)$ denotes the $k_1k_2{}^{th}$ pair in $\mathcal{S}_i$. With the initial value ϕ^0, α^{K+1} can be estimated using the ordinary least squares (OLS) estimator as follows:

$$\hat{\alpha}_0^{K+1} = \left(\sum_{i=1}^{N} \tilde{\mathcal{Z}}_i^T\tilde{\mathcal{Z}}_i\right)^{-1}\left(\sum_{i=1}^{N} \tilde{\mathcal{Z}}_i^T\tilde{U}_i\right).$$

With $\hat{\alpha}_0^{K+1}$, the variance parameter $\phi_{k_1k_2}$ can be updated by solving the following weighted estimating equation:

$$\sum_{i=1}^{N} \frac{R_i^{k_1}R_i^{k_2}\left((U_i^{k_1k_2} - A^{k_1k_2}Z_i\hat{\alpha}_0^{K+1})^2 - \phi_{k_1k_2}\right)}{\left(1-\exp(-\eta Y_i^{k_1})\right)\left(1-\exp(-\eta Y_i^{k_2})\right)} = 0,$$

which gives the solution

$$\phi_{k_1k_2}^1 = \left(\sum_{i=1}^{N} \frac{R_i^{k_1}R_i^{k_2}(U_i^{k_1k_2} - A^{k_1k_2}Z_i\alpha_1^{K+1})^2}{\left(1-\exp(-\eta Y_i^{k_1})\right)\left(1-\exp(-\eta Y_i^{k_2})\right)}\right) / \left(\sum_{i=1}^{N} \frac{R_i^{k_1}R_i^{k_2}}{\left(1-\exp(-\eta Y_i^{k_1})\right)\left(1-\exp(-\eta Y_i^{k_2})\right)}\right).$$

Let $\phi^1 = (\phi_{12}^1, \phi_{13}^1, \cdots, \phi_{K,K+1}^1)$. With this, the value of α_0^{K+1} can be updated to α_1^{K+1} using the OLS estimate. We can continue with this iteration until convergence (e.g., $||\hat{\alpha}_1^{K+1} - \hat{\alpha}_0^{K+1}||_1 < 10^{-8}$ and $||\phi^1 - \phi^0||_1 < 10^{-8}$).

4.1.2 Determine η with cross validation

In practice, Y_i^k is a count variable which only takes a positive integer or zero, so we assume $Y_i^k = [\mathcal{Y}_i^k \cdot C_i]$ where $[\cdot]$ denotes the integer part. Notice that

$$\begin{aligned} P(R_i^k = 0) &= \sum_{j=0}^{\infty} P(R_i^k = 0, Y_i^k = j) \\ &= P(R_i^k = 0, Y_i^k = 0) + \sum_{j=1}^{\infty} P(R_i^k = 0|Y_i^k = j)P(Y_i^k = j) \\ &= P(R_i^k = 0, Y_i^k = 0) + \sum_{j=1}^{\infty} \exp(-\eta j) \cdot P(Y_i^k = j), \end{aligned}$$

where the two terms of the right-hand side represent floor zero (the first part) and mechanical zero, respectively. And by using the conditional probability formula, the probability $P(Y_i^k = j)$ can be written as:

$$P(Y_i^k = j) = \frac{P(Y_i^k = j, R_i^k = 1)}{P(R_i^k = 1|Y_i^k = j)} = \frac{P(Y_i^k = j, R_i^k = 1)}{1 - \exp(-\eta j)}.$$

After plugging this into the first equation above, we have

$$\begin{aligned} P(R_i^k = 0) &= P(R_i^k = 0, Y_i^k = 0) + \sum_{j=1}^{\infty} \frac{\exp(-\eta j) P(Y_i^k = j, R_i^k = 1)}{1 - \exp(-\eta j)} \\ &= P(R_i^k = 0, Y_i^k = 0) + \sum_{j=1}^{\infty} \frac{P(Y_i^k = j, R_i^k = 1)}{\exp(\eta j) - 1}. \end{aligned}$$

It is straightforward to see that the right-hand side of the above equation is a decreasing function of η which implies there is a unique solution for η. Notice that $P(R_i^k = 0)$ can be estimated by the proportion of observed zeros in the data, and $P(Y_i^k = j, R_i^k = 1)$ can be estimated by the proportion of observed values of j in the data. Thus an estimate of η can be obtained by solving the above equation with those observed proportions if the proportion of mechanical zero is given. A k-fold cross-validation can be used to determine the final value η:

1. Choose 11 candidate values $\eta_1, \cdots, \eta_{10}$ corresponding to $0\%, 10\%, \cdots, 90\%, 100\%$ of observed zeros being mechanical zeros respectively. Let p denote the percentage, then η can be solved from $p \times P(R_i^k = 0) = \sum_{j=1}^{\infty} \frac{P(Y_i^k = j, R_i^k = 1)}{\exp(\eta j) - 1}$ for each p. The number of candidate values for η does not have to be 11, and it can be customized for different studies.

2. For each candidate value, run a typical 5-fold (or other k-fold) cross-validation and find the summation of mean square error (MSE) over all the testing partitions. The MSE is defined as $MSE^{\xi} = \sum_{1 \le k_1 < k_2 \le K+1} \frac{1}{|N_\xi|} \sum_{i \in N_\xi} \left(R_i^{k_1} R_i^{k_2} \phi_{k_1 k_2}^{-1} (U_i^{k_1 k_2} - A^{k_1 k_2} \mathcal{Z}_i \alpha^{K+1})^2 \right)$, $\xi = 1, \cdots, 5$, where N_ξ denotes the ξ-th testing partition and $|N_\xi|$ denotes the sample size in that partition.

3. The final value of η is chosen to minimize the testing MSE which is calculated as $CV(\eta) = \frac{1}{5} \sum_{\xi=1}^{5} MSE^{\xi}$.

The cross-validation process is conducted independently from the estimation part, which reflects the optimal practice in operational use. Therefore, the procedure serves as a integral part in the model estimation. The validation step reduces the selection bias in η and yields to robust variance estimates (Cawley and Talbot (2010)).

4.1.3 Variance estimation and inference

The estimators given by the weighted estimation equation are asymptotically normal (Robins et al. (1995)), but solving the sandwich variance estimators is still expensive in computation. We propose to estimate the variance using Jackknife method as it is a common method used in

literature for correlated data (Shao and Tu (2012)). Delete-one jackknife estimators are typically used. Let $\hat{\alpha}_{(i)}^{K+1}$ denote the delete-one estimate obtained by running the weighted estimating equation on the subset after deleting the ith subject, $i = 1, \cdots, N$. A jackknife estimate, $\hat{\alpha}_J^{K+1}$, can be calculated as follows:

$$\hat{\alpha}_J^{K+1} = \hat{\alpha}^{K+1} - \frac{N-1}{N} \sum_{i=1}^{N} (\hat{\alpha}_{(i)}^{K+1} - \hat{\alpha}^{K+1}),$$

where $\hat{\alpha}^{K+1}$ is the full data estimate. The estimated variance of $\hat{\alpha}_J^{K+1}$ is calculated as:

$$\widehat{\text{Var}}(\hat{\alpha}_J^{K+1}) = \frac{N-1}{N} \sum_{i=1}^{N} (\hat{\alpha}_{(i)}^{K+1} - \hat{\alpha}_{(\cdot)}^{K+1})(\hat{\alpha}_{(i)}^{K+1} - \hat{\alpha}_{(\cdot)}^{K+1})^T,$$

where $\hat{\alpha}_{(\cdot)}^{K+1} = \frac{1}{N} \sum_{i=1}^{N} \hat{\alpha}_{(i)}^{K+1}$. With the estimated variance, standard errors can be calculated and confidence intervals and p-values can be generated for making statistical inference. For studies with large sample sizes, block-wise Jackknife estimators can be used to reduce computational burden. The jackknife estimator for variance is approximately unbiased given proper selection of the block size (Kott (1998)). Therefore, it can be used as an alternative to the sandwich variance estimator.

4.2 Unknown reference taxon

In applications, the taxa that are independent of the X are generally unknown. Following the strategy of IFAA (Li et al. (2021)), we identify putative reference taxa by examining how often each taxon is selected across analyses conducted with different reference taxa. Partition the $K+1$ taxa into two unknown groups. Let A contain taxa associated with X, and let B contain taxa whose associations with X are absent. For taxa in B, $\beta^k = \mathbf{0}$; consequently, the log-ratio between any two taxa in B is independent of X. In contrast, log-ratios involving two taxa in A, or one taxon from each group, remain associated with the covariates. This leads to two cases depending on the reference taxon. If the reference belongs to B, the procedure should identify every taxon in A as associated with X, while taxa in B should not be selected. If the reference belongs to A, then all other taxa including those in both A and B should be selected. Denote the cardinalities of the two groups by m_A and m_B, so that

$$m_A + m_B = K + 1.$$

If each taxon were used once as the reference, then a taxon in B would be selected m_A times, whereas a taxon in A would be selected K times. Thus, provided $m_B > 1$, the two groups could be distinguished from their selection frequencies, since

$$K - m_A = m_B - 1 > 0$$

To reduce computational burden, we randomly sample R reference taxa, for example $R = 20$, and apply the procedure separately for each sampled reference. Under this scheme, the expected

number of selections for a taxon in A and B are:

$$k_A = \frac{RK}{K+1}, k_B = \frac{Rm_A}{K+1}.$$

The difference between the expected selection frequencies is

$$k_A - k_B = \frac{KR - Rm_A}{K+1} = \frac{R(m_B - 1)}{K+1} > 0,$$

assuming $m_B > 1$. Therefore, taxa with comparatively low selection frequencies are expected to belong to B, making them natural candidates for use as reference taxa. In the case that there are taxa in set A having the same association strength with X, we have $k_A \geq Rm_B$. Thus, as long as $m_B > m_A$, the difference $k_A - k_B \geq (m_B - m_A) > 0$ means set B can still be identified.

5 Simulation

Extensive simulations were conducted to demonstrate the performance of the proposed method compared with an existing popular method in the literature MaAsLin2 (Mallick et al. (2021)). We consider two settings: the first setting is differential abundance analyses with a binary covariate and the second setting is regression analyses with a continuous covariate.

5.1 Simulation setting one

In this section, the covariate of interest (i.e. X_i) is binary to demonstrate performance on differential abundance analyses. 50 random datasets, each having 100 subjects and 400 taxa were generated. 25% of the taxa were assumed to be differentially abundant, and the two-group indicator variable X_i was generated using a Bernoulli distribution with a parameter value of 0.5.

The true mean absolute abundance in the ecosystem $\mathcal{Y}_i^k$ was generated according to model (1). Random Gaussian noise was added to the right hand side of model (1). For those differentially abundant taxa, the β^k were generated such that 40% were having low effect size generated by the uniform distribution $U(1.39, 2.08)$ and 60% were having high effect size generated by $U(2.77, 3.47)$. The β^k of all other taxa were set to be 0. A random intercept was included for each sample to account for within-sample correlation and it was generated by a normal distribution with mean 0 and variance 0.5. The variance of the normally distributed error was independently generated from $U(3, 4)$. Under such model parameters, the signal-to-noise ratios (SNR) in the datasets were between 0.95 and 0.99.

The sample absolute abundance Y_i^k were generated by $Y_i^k = [C_i \mathcal{Y}_i^k]$ where $[\cdot]$ will induce floor zeros space. The mechanical zeros were generated according to equation (2).

The performance of association identification was measured by four indices: Recall, Precision, F1, and Type I error rate (Type1). They were calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{F1} = \frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}, \text{Type1} = \frac{\text{FP}}{\text{TN} + \text{FP}}$$

We studied five scenarios where the ratios of the latent variable C_i between the groups, $C_i(X_i = 0)/C_i(X_i = 1)$, were set to be 1, 3, 5, 7, and 9 respectively. In all scenarios, the overall sparsity level was 50%, with half mechanical zeros and half floor zeros. Figure 1 shows that IWEE remains highly stable across all imbalance levels: precision stays between 0.98 and 0.99, recall is consistently between 0.82 and 0.87, and F1 score remains near 0.89 and 0.91, while Type I error is low in all cases. In contrast, MaAsLin2 deteriorates sharply as the imbalance increases: precision drops from about 0.88 at a ratio of 1 to 0.25 for ratios ≥ 3. Recall decreases from around 0.90 to 0.71; and F1 score declines from 0.92 to 0.30. Moreover, its Type I error inflates dramatically from about 0.05 at a ratio of 1 to 0.90-1.00 for ratio ≥ 3. These results demonstrate that IWEE is robust to sequencing depth imbalance, while MaAsLin2 becomes unreliable with substantial loss of accuracy and extreme false positive rates.

We also studied the effect of sparsity level. The overall sparsity levels were set to 10%, 20%, 30%, 40%, 50% and 60% with half mechanical zero and half floor zero. The sequencing depth was set such that $C_i(X_i = 0)/C_i(X_i = 0) = 1) = 1.3$ in all sparsity levels. The results are shown in Figure 2. As sparsity increases from 10% to 60%, IWEE maintains near perfect precision (around $0.98 - 0.99$). There is a decline in recall from about 0.95 to 0.70 as overall sparsity increases from 10% to 60%, leading to a gradual decrease in F1 score from 0.96 to 0.82; Type I error remains controlled at a low level throughout. In comparison, MaAsLin2 shows a more variable performance: precision fluctuates between 0.68 and 0.78, recall remains relatively high (around 0.96–0.99), and F1 score stays around 0.82–0.88. However, its Type I error is consistently inflated, ranging from about 0.10 to 0.18. Overall, IWEE provides more reliable inference with strict error control, while MaAsLin2 trades higher recall for substantially increased false positives.

Regarding the effect of floor zero proportion, we consider the scenarios with floor zero percentage among all zeros in the dataset to be 100%, 75%, 60%, 45%, 30%, and 15% respectively, with 50% overall sparsity level and sequencing depth ratio $C_i(X_i = 0)/C_i(X_i = 1)) = 1.3$. The results are shown in Figure 3. IWEE maintains high precision across all floor zero proportions and keeps Type I error essentially at zero. Its recall is lower at small floor zero shares but increases to 1.0 when all zeros are floor zeros; consequently, its F1 rises and peaks at 100% floor zero. MaAsLin2 shows uniformly high recall (almost 1.0) regardless of floor zero proportion. Its precision declines as floor zeros increase and its Type I error inflates, especially at 100% floor zero (largest error), leading to a drop in F1. At moderate floor zero levels (30%–60%), the methods have similar F1, reflecting a tradeoff that MaAsLin2 favors sensitivity at the expense of specificity, whereas IWEE provides stronger error control and becomes clearly superior when floor zeros dominate.

5.2 Simulation setting two

In this section, we demonstrate the proposed method in regression analysis settings with a continuous covariate of interest (i.e. X_i). We generated 50 random datasets with each having 100

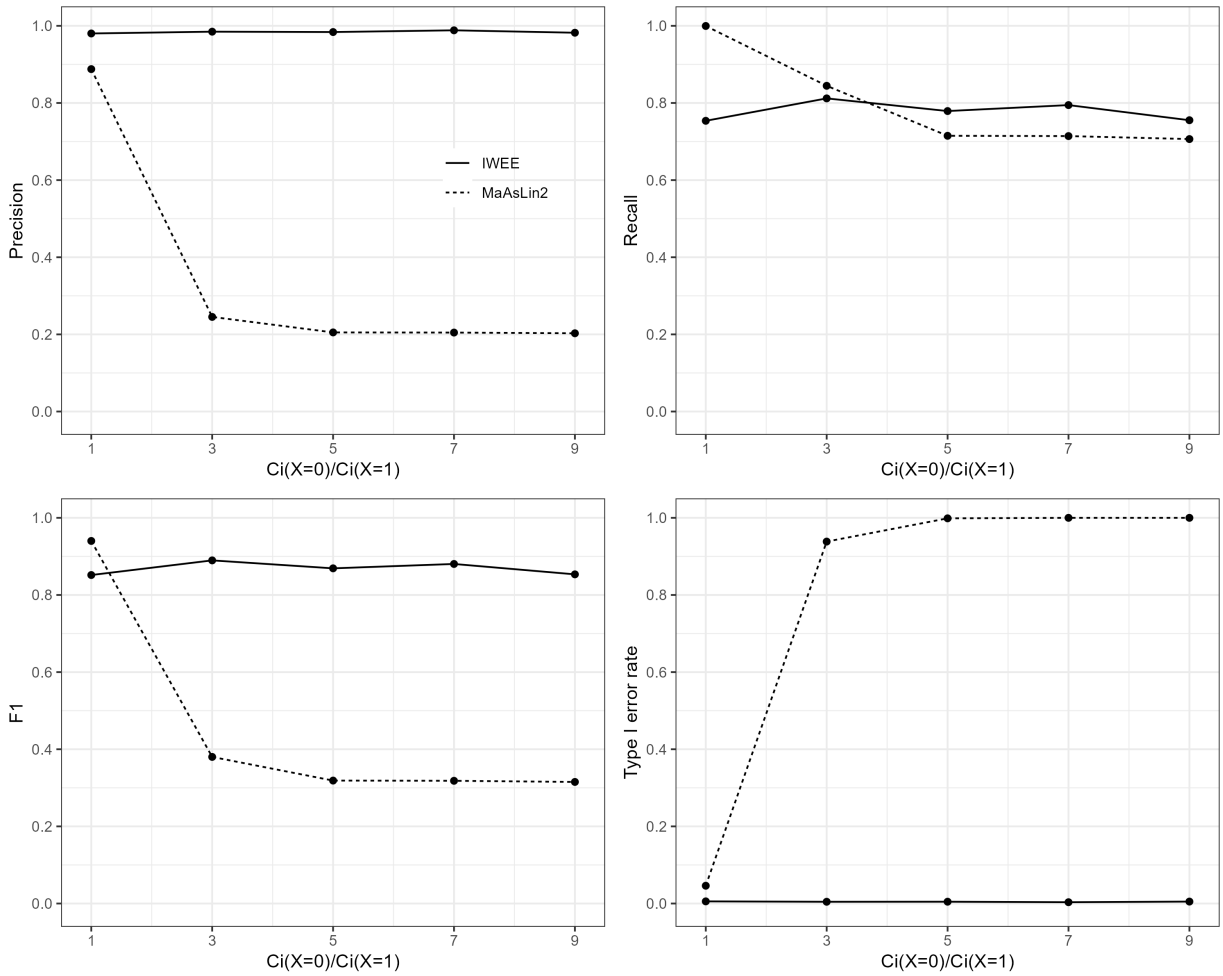


Figure 1: Comparison with MaAsLin2. The effect of unbalanced sequencing depth in differential abundance analysis.

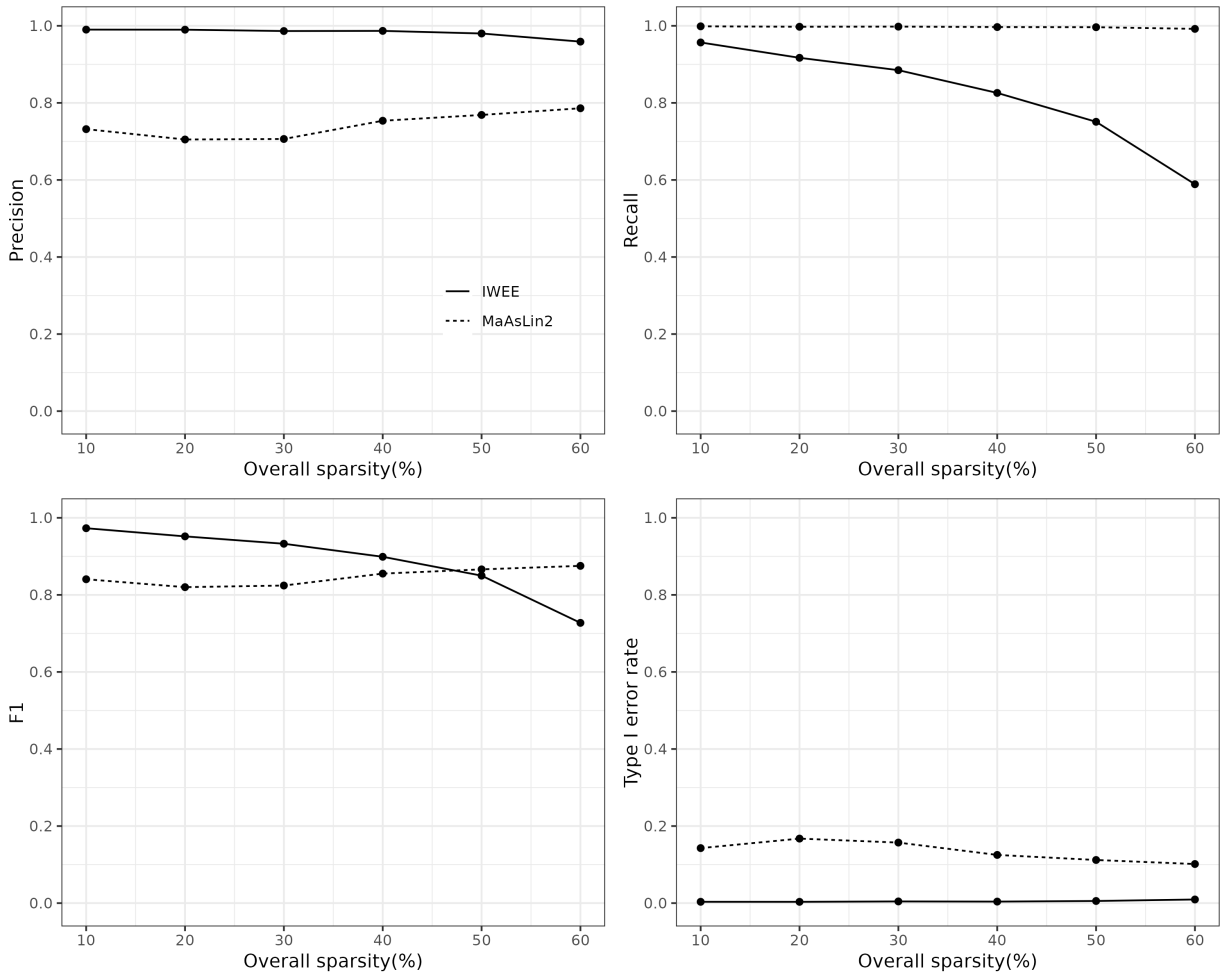


Figure 2: Comparison with MaAsLin2. The effect of overall sparsity level in differential abundance analysis.

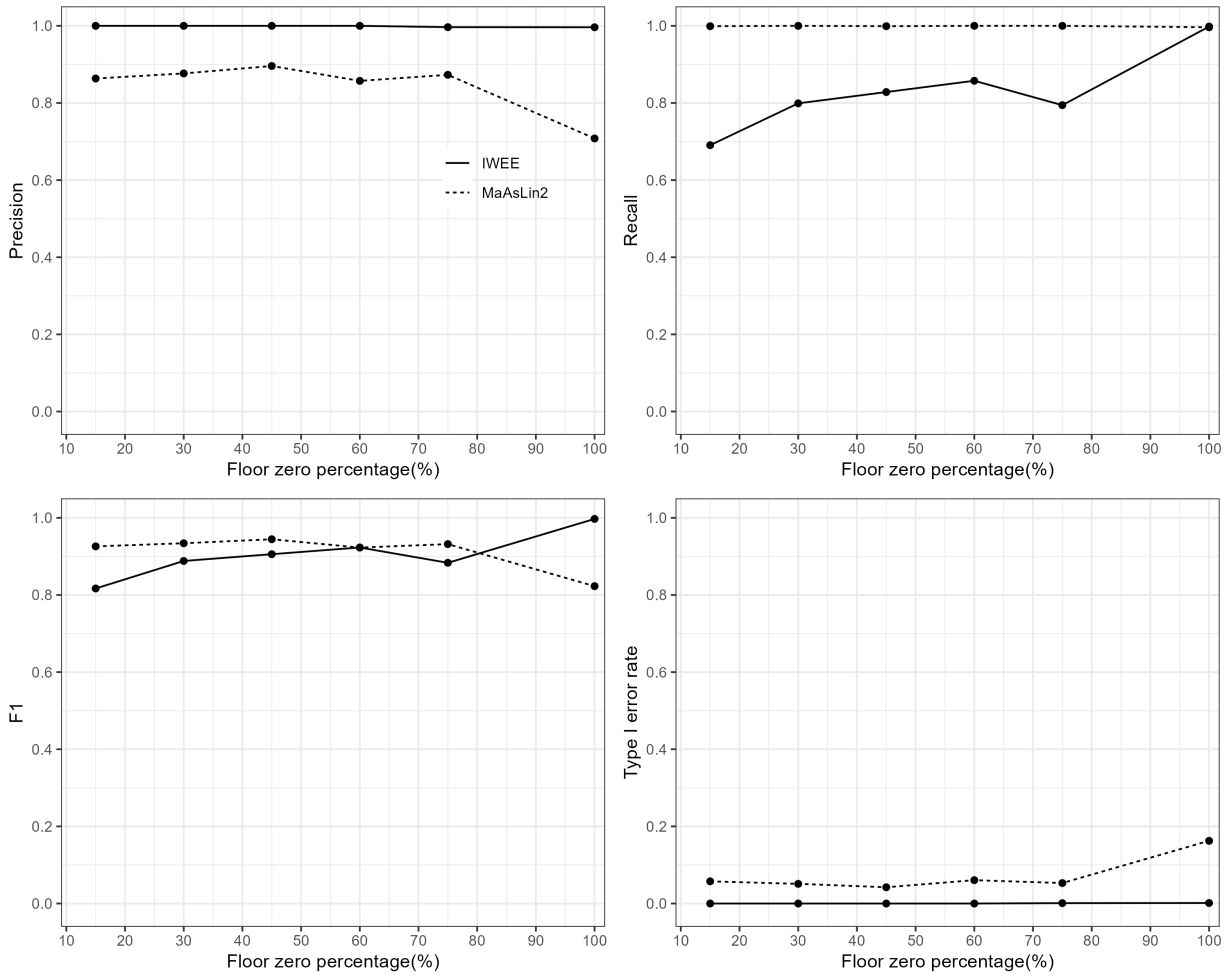


Figure 3: Comparison with MaAsLin2. The effect of proportion of floor zeros in differential abundance analysis.

subjects, and 400 taxa. 25% of the taxa were assumed to be significantly associated with covariate of interest. We generated X from $U(-0.5, 1)$. The effect size for significant taxa were set as, similar to Section 5.1, 40% were low from $U(2.08, 2.77)$, and 60% were high from $U(2.77, 3.47)$. The variance of normal error were generated from $U(2, 3)$ and the random intercept was generated from $\sqrt{0.5}N(0, 1)$. As a result, the SNR were between 1 and 1.2. C_i was generated as follows:

$$C_i = \max\left(\frac{1}{1000}, \min\left(sigmoid(-10X_i + 0.1N(0,1))/100, \frac{1}{100}\right)\right),$$

where $sigmoid(x) = \frac{1}{1+exp(-x)}$. Notice that C_i is bounded between $\frac{1}{1000}$ and $\frac{1}{100}$ and is dependent on X_i in this setting.

Performance was studied under overall sparsity levels of $10\%, 20\%, 30\%, 40\%, 50\%, 60\%$ with half mechanical zeros and half floor zeros. The results were plotted in Figure 4. Across overall sparsity levels, IWEE maintains high precision and controls Type I error near zero. Recall stays above 0.9 through 50% sparsity but declines at 60% to 60%. Therefore, the F1 remains near 1.0 up to 50% and drops modestly at 60% to 0.75. In contrast, MaAsLin2 shows low precision and an inflated Type I error. The precision drops from 0.39 to 0.28 and the Type I error increases from 0.64 to almost 1 as the overall sparsity level increases from 10% to 60%. While its recall is around 1.0, the F1 of MaAsLin2 remains low.

We also studied the impact of floor zeros. As shown in Figure 5, across all floor zero settings, IWEE again shows stable performance: precision remains between 0.98 and 0.99, recall is consistently high at $0.92-0.96$, and F1 score stays in the range of $0.95-0.97$, with Type I error equal to 0. In contrast, the precision of MaAsLin2 is low, which is between 0.25 and 0.40 for most settings. The F1 score ranges from 0.40 to 0.60 although recall remains high (around $0.95-0.99$) at the cost of high Type I error, which is approximately 1.00 for 30%–75% floor zeros, and still above 0.50 at other floor zero proportion settings. These results indicate that IWEE is robust to the composition of zero types and outperforms MaAsLin2.

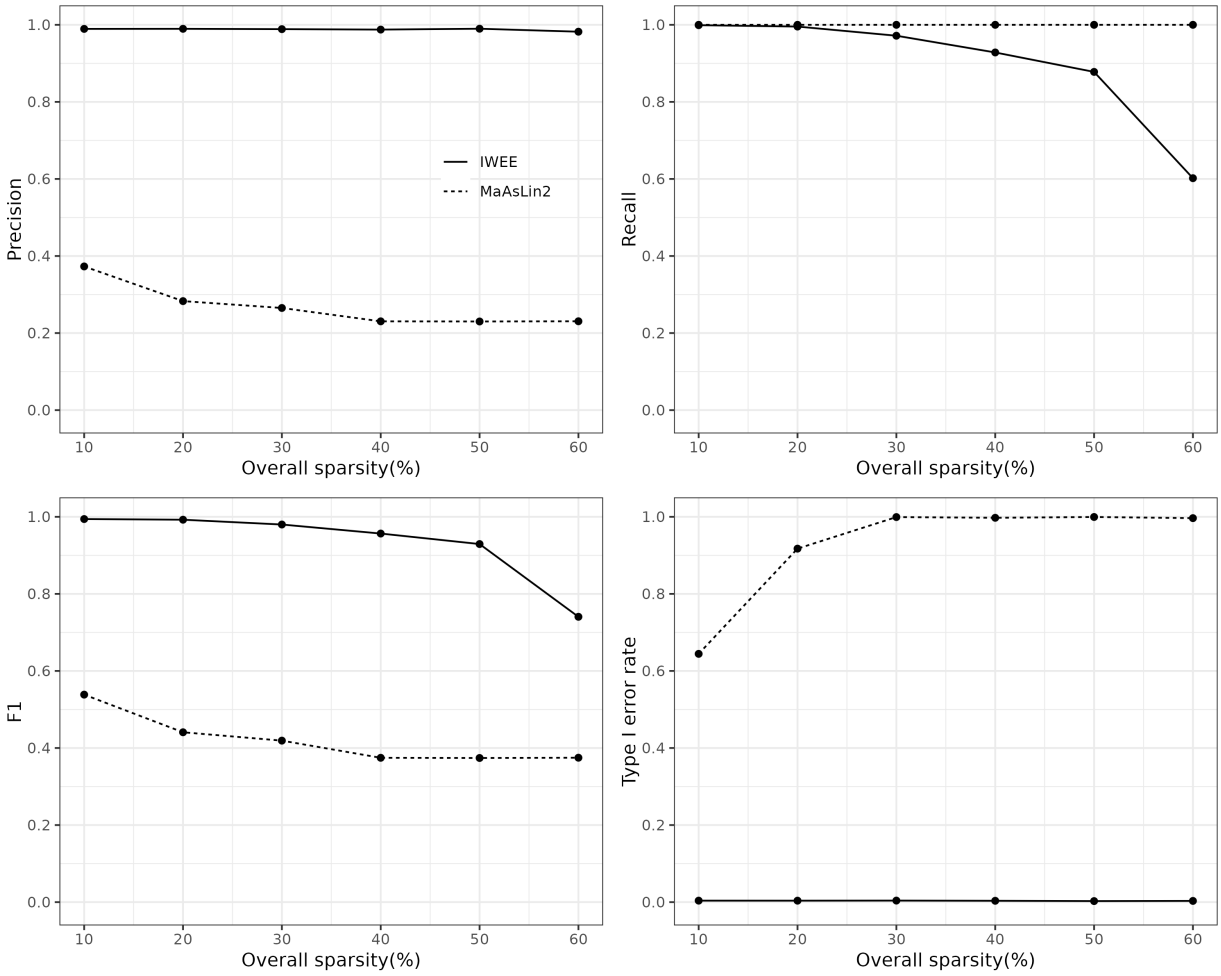


Figure 4: Comparison with MaAsLin2. The effect of overall sparsity level in regression analysis.

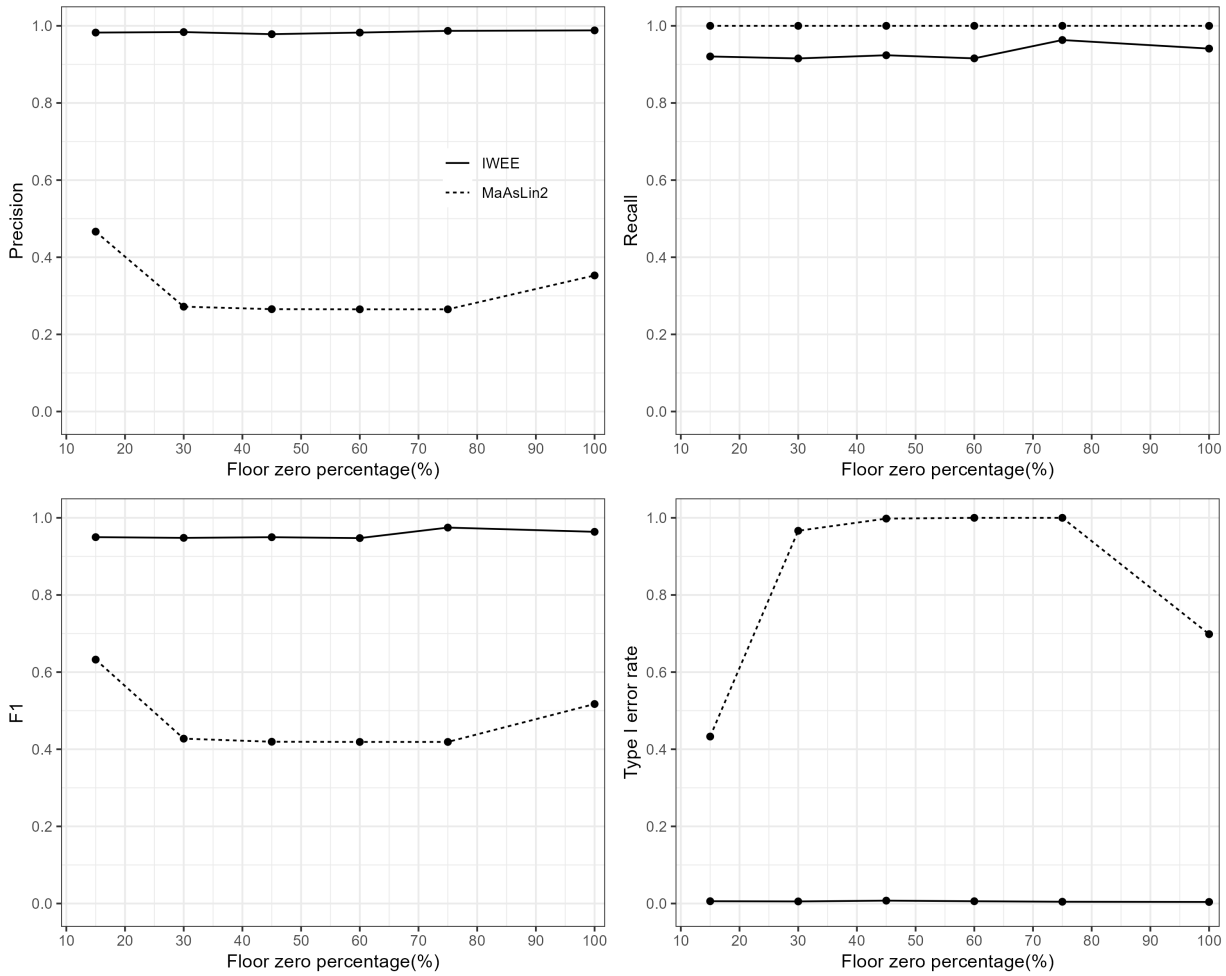


Figure 5: Comparison with MaAsLin2. The effect of proportion of floor zeros in regression analysis.

6 Real data application

6.1 Marijuana Associated Planning and Long-term Effects Study

The Marijuana Associated Planning and Long-term Effects (MAPLE) study is a longitudinal cohort designed to investigate the impact of marijuana (cannabis) use on health outcomes over time among individuals living with HIV (Porchia et al. (2025)). In this analysis, we focused on the Alzheimer's disease supplemental sub-study, which includes 64 participants diagnosed with mild cognitive impairment (MCI). Although MAPLE is longitudinal in design, the present analysis is cross-sectional and uses baseline measurements of marijuana exposure and microbiome composition generated from 16S rRNA sequencing technology. Following quality control and data cleaning, which includes the removal of low-quality sequencing reads, potential contaminants, and features with insufficient abundance or prevalence, and excludes samples or genera with inadequate data coverage, the final 16S rRNA sequencing dataset included 248 genera for regression analysis.

We applied the proposed method to examine the association between gut microbiome composition and marijuana consumption. The outcome variable Y in Section 3 corresponds to genus-level gut microbiome data. For each genus, sample absolute abundance (AA) was used as the outcome measure. Prior to the analysis, seven taxa with no sequencing reads were removed. The primary covariate of interest, X, was baseline marijuana use, quantified in grams per day (Mean = 107, Median = 56, SD = 164, Interquartile Range = (0, 148)). Participant age and gender were included as potential confounders in W. The overall sparsity of the microbiome data was approximately 73%. The tuning parameter η, selected by cross-validation, was estimated to be ∞, indicating that all zero observations were treated as floor zeros.

After applying the IWEE method, 30 genera were identified as significantly associated with marijuana consumption, of which 3 remained significant after FDR adjustment for multiple comparisons. The association heatmap in Figure 6 illustrates the estimated effects across genera. Using MaAsLin2, 14 genera were identified as significantly associated with marijuana consumption, with 3 remaining significant after adjustment for multiple comparisons.

Among the genera identified as significant by IWEE, *Isoptericola* ($\beta = -0.29, FDR = 0.02$) has been linked to several human diseases. Its abundance in the gut and blood has been associated with Parkinson's disease and acute lymphoblastic leukemia (Muratore et al. (2025), Qian et al. (2018)). *Leptotrichia* ($\beta = -0.23, FDR = 0.04$) has been reported to occur at higher relative abundance in tumor tissue than in non-tumor tissue (González et al. (2024)) and has also been associated with cardiovascular disease in patients with colorectal cancer (Druzhinin et al. (2026)). The relevance of *Basfia* ($\beta = 0.26, FDR < 0.001$) to human health remains unclear, as this genus has primarily been studied in the context of industrial biotechnology (Arshad et al. (2025), Lange et al. (2017)). MaAsLin2 identified three significant taxa. Previous studies have associated *Victivallaceae* ($\beta = 0.09, FDR < 0.001$) with celiac disease (Zou et al. (2024)) and tumor recurrence in patients with head and neck cancer (Al-Qadami et al. (2023)). The abundance of *Marvinbryantia* ($\beta = 0.21, FDR = 0.002$) may increase in response to a high-fat diet and has been linked to liver

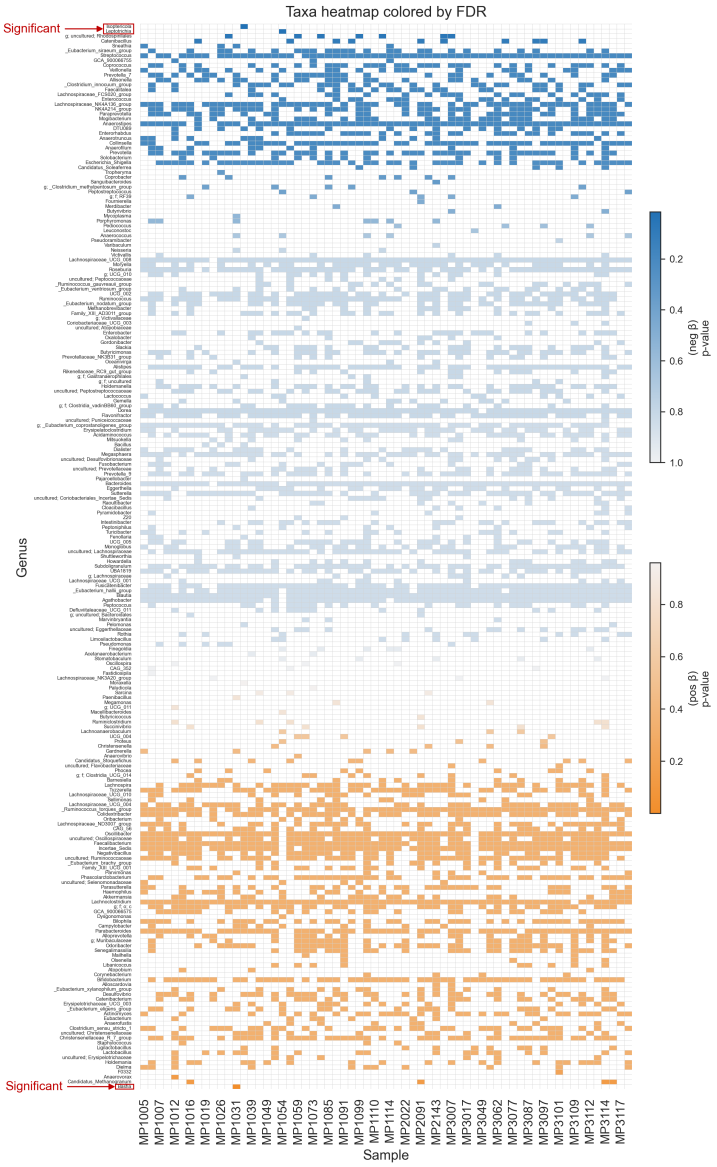


Figure 6: Association heatmap of genera identified by IWEE in the MAPLE study.

cirrhosis (Yuan et al. (2024)). Finally, the potential effects of *Mailhella* ($\beta = 0.24, FDR < 0.001$) on human health remain uncertain because it is a relatively recently characterized genus (Schoch et al. (2020)).

6.2 Liberalized Diet and Neutropenic Diet Study

We applied the IWEE method to microbiome data from participants enrolled in a noninferiority Phase III clinical trial comparing a liberalized diet with a neutropenic diet among hematopoietic stem-cell transplant recipients or remission induction chemotherapy for acute leukemia and myelodysplastic sybdrome (Wingard et al. (2026)). We focused on the end of intervention time point to evaluate the association between dietary assignment and microbiome composition as a

secondary aim. After quality control and data cleaning for omics data, the 16S rRNA sequencing dataset comprised 88 subjects and 304 bacterial genera for differential abundance analysis.

The outcome variable Y_i in Equation 1 corresponded to genus level microbiome data derived from raw ASV counts. The resulting genus-level abundance data were then matched to participant metadata and treatment assignment. The primary covariate of interest, X_i, was a binary variable denoting randomized diet group. No additional control covariate, W_i, was included in the model. We then used IWEE to assess the association between diet assignment and microbiome composition, and the resulting estimates were used for downstream inference. Neutropenic diet was selected as the reference group.

After applying the IWEE method, 46 genera were identified as significantly associated with diet group. After FDR correction for multiple comparisons, 4 genera remained significantly associated with diet group: *Scardovia* ($\beta = 5.401, FDR = 0.024$), *Streptococcus* ($\beta = 2.60, FDR = 0.034$), *Asaccharospora* ($\beta = 2.368, FDR = 0.016$), and *Romboutsia* ($\beta = 1.638, FDR = 0.016$. Figure 7 visualizes the difference in the average RA between diet groups for those significant genera. The association heatmap in Figure 8 illustrates the estimated effects across genera. Using MaAsLin2, 9 genera were identified as significantly associated with diet group, but none remained significant after FDR correction.

Previous studies have shown that *Streptococcus* is inversely associated with dietary fructose intake (Jones et al. (2019)). It is also inversely related to the sphingosine and "Glycosphingolipid biosynthesis" pathway among high-fat diet subjects (Wei et al. (2021)). Many studies identified that *Roboutsia* composition in oral and gut ecosystem are enriched by high-fat diet in all age groups (Liu et al. (2022), Tian et al. (2026), Yin et al. (2023)). Researchers also revealed that *Scardovia* is associated with dental caries (Wang et al. (2025)) and can ferment arbohydrates (Henne et al. (2015)). *Asaccharospora* can be enriched by treating with flavonoids of different structures (Mao et al. (2024)).

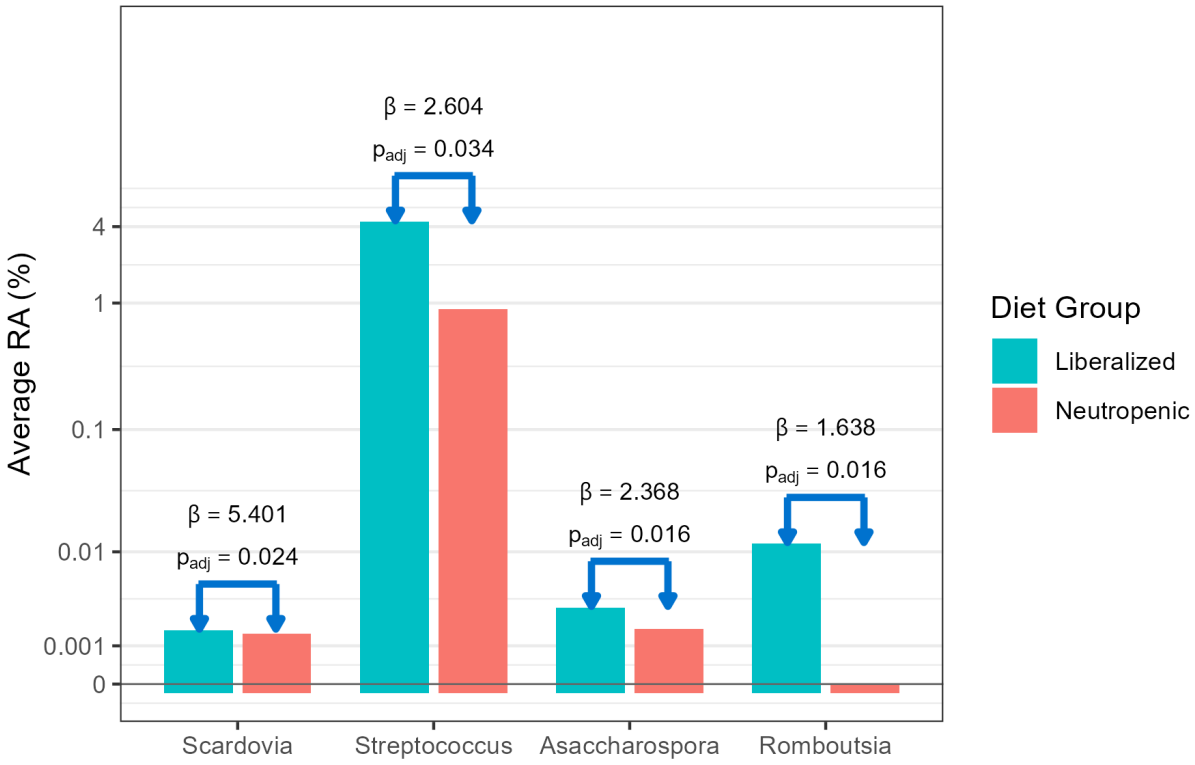


Figure 7: Average relative abundance (RA) for significant genus in the Liberalized and Neutropenic Diet groups. Beta is reported with the Neutropenic Diet as the reference group.

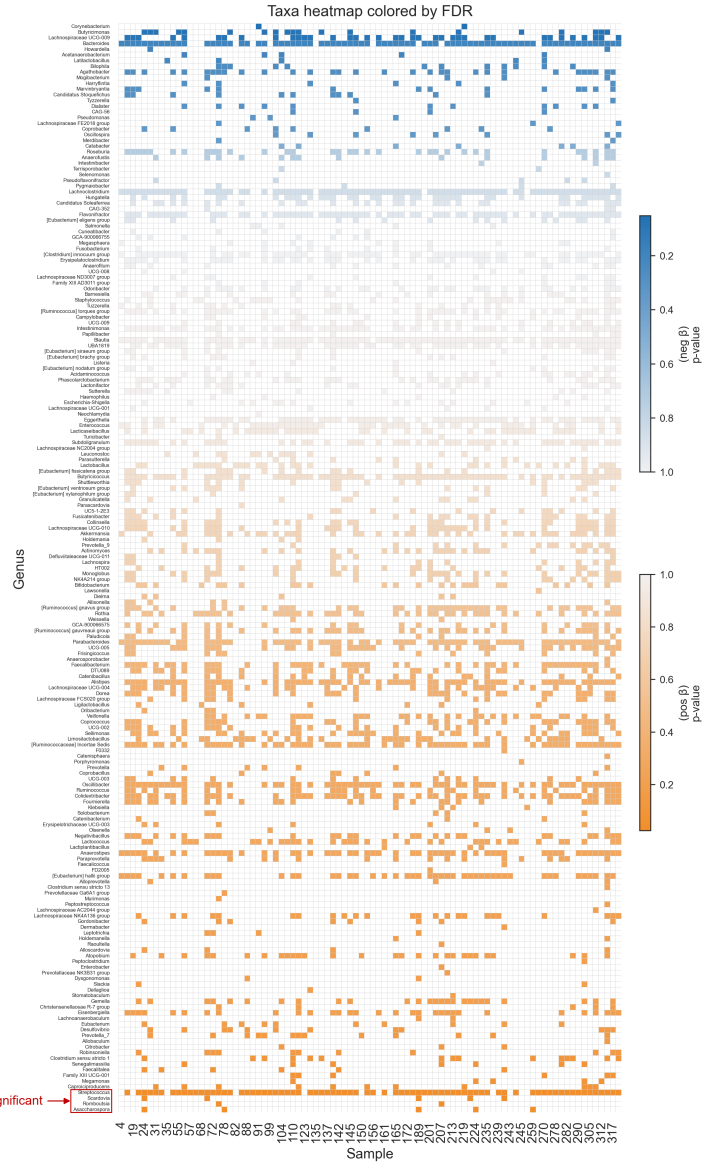


Figure 8: Association heatmap of genera identified by IWEE in the Neutropenic Diet study.

7 Discussion

We developed a novel approach (IWEE) to account for excessive zeros in microbiome data analysis including differential abundance analysis and regression analysis by employing weighted estimating equations. Similar to inverse probability weighting for dealing with missing data, zeros were treated as missing data and the probability of observing a mechanical zero is modeled by a probability mechanism. The zero mechanism includes both mechanical zeros and floor zeros, where floor zeros can include structural zeros. K-fold cross-validation was used to estimate the parameter in the probability mechanism. A weighted estimating equation was constructed for estimation on the coefficients, and the Jackknife method was used to obtain their standard errors. In addition, to address compositional structure, the idea of the IFAA method was applied to identify reference taxa that are independent of the covariates of interest, and then estimate the parameters of interest based on the identified reference taxa. This new approach makes inference on AA and can provide consistent estimators under the mechanical zero mechanism. It also has a superior performance in comparison with MaAsLin2 when there is a mix of mechanical zeros and floor zeros, as shown in the extensive simulations under various scenarios, such as uneven sequencing depths and different sparsity levels. Another advantage of using estimating equations is that there is no distributional assumptions on the data, so that the model is robust to real world distributions and can appropriately account for inter-taxa correlations. Potential confounders can be adjusted in the approach and FDR adjusted p-values can be generated in the results. The approach can be applied to both RA data and AA data, although the final inference is on AA in the ecosystem. An R package is available on GitHub to implement the method (`https://github.com/TChang-W/IWEE-microbiome`).

The current implementation of IWEE adjusts p-values using the Benjamini–Hochberg procedure to control the false discovery rate (FDR). Other procedures may be preferable depending on the inferential goal, dependence among tests, and desired balance between false-positive control and statistical power. For example, the Benjamini–Yekutieli procedure controls the FDR under more general dependence conditions Benjamini and Yekutieli (2001), while Storey's q-value method estimates the proportion of true null hypotheses and can improve power in some settings Storey (2002). Adaptive procedures, including the adaptive resampling test McKeague and Qian (2015), offer another approach to identifying significant predictors. More recent methods also provide flexible FDR control: IHW incorporates covariate information to improve power while controlling the FDR Ignatiadis et al. (2016), and AdaPT uses a data-adaptive, interactive procedure to leverage covariates while maintaining FDR control Lei and Fithian (2018). Comparing these alternatives and evaluating their performance under the dependence structures and sparsity patterns encountered in microbiome studies are important directions for future work.

The current version of IWEE does not allow high-dimensional covariates in X_i (i.e. $Q > N$). A possible extension can be made by employing high-dimensional regression methods using estimating equations (Neykov et al. (2018), Robins et al. (2017), Wang et al. (2011)). Another possible extension is to adapt a different probability mechanisms for the mechanical zeros and

identify the optimal mechanisms with cross-validation.

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